

ORENBURG, Russian Federation

WMO: 351210

Lat: 51.6925N

Lon: 55.0783E

Elev: 386

StdP: 14.49

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.1	-15.2	-27.2	1.2	-21.0	-21.0	1.8	-14.8	26.1	15.1	23.2	14.7	3.8	80	0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.9	94.7	66.9	91.3	66.1	87.8	65.2	70.4	86.2	68.7	84.6	67.1	82.7	9.5	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.3	94.8	76.1	63.1	87.8	74.3	61.4	82.3	73.1	34.6	86.2	33.1	84.2	31.8	82.2	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.0	20.4	18.2	-28.8	99.5	6.0	3.4	-33.1	101.9	-36.6	103.9	-40.0	105.8	-44.4	108.3		
(5)			-29.0	73.5	6.0	2.6	-33.3	75.4	-36.8	76.9	-40.2	78.4	-44.5	80.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.6	8.9	10.6	23.3	45.0	59.3	68.1	71.4	69.2	57.2	43.0	27.4	14.4		
	DBStd	24.89	12.97	12.71	10.30	10.89	8.38	8.12	6.49	7.87	8.36	8.52	11.20	13.06		
	HDD50	5506	1273	1103	829	221	19	2	0	28	251	678	1102			
	HDD65	9134	1738	1523	1294	602	217	59	19	47	257	684	1127	1567		
	CDD50	2457	0	0	0	71	306	544	664	595	244	33	0	0		
	CDD65	609	0	0	0	2	39	151	218	176	23	0	0	0		
	CDH74	7924	0	0	0	56	642	1941	2605	2271	404	5	0	0		
	CDH80	3386	0	0	0	6	194	861	1134	1064	127	0	0	0		
	WSAvg	8.5	8.6	8.5	9.3	9.4	9.1	8.3	8.0	7.8	7.5	8.2	8.5	8.4		
	PrecAvg	14.4	1.1	0.8	0.8	0.9	1.1	1.6	1.7	1.1	1.1	1.4	1.3	1.3		
(15) Precipitation	PrecMax	21.1	3.1	2.2	2.3	2.5	3.1	4.6	5.3	3.3	3.5	3.9	3.3	2.8		
	PrecMin	8.5	0.2	0.0	0.0	0.0	0.1	0.0	0.0	0.1	0.1	0.1	0.2	0.2		
	PrecStd	3.2	0.6	0.5	0.6	0.7	0.7	1.0	1.2	0.8	0.8	0.8	0.7	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.2	35.3	53.4	80.3	88.8	97.3	96.7	98.6	89.9	73.3	52.2	37.5		
		MCWB	34.3	33.7	44.7	58.7	63.3	67.1	68.1	67.1	61.4	55.2	47.4	35.0		
	2%	DB	32.4	33.6	41.3	74.9	85.4	93.3	93.2	94.8	83.1	67.7	47.1	34.4		
		MCWB	31.6	32.8	37.7	56.3	62.1	66.8	67.3	66.3	60.3	52.4	44.0	33.3		
	5%	DB	30.5	31.9	37.7	68.9	81.6	89.2	90.0	91.0	78.6	62.4	43.5	33.5		
		MCWB	29.8	31.0	35.1	53.0	60.7	65.5	66.7	65.2	59.1	49.5	41.2	32.6		
	10%	DB	28.0	28.8	35.7	63.5	76.9	84.7	86.5	86.1	73.5	57.1	40.7	31.8		
		MCWB	27.1	27.7	33.7	50.3	58.8	64.4	66.2	64.5	57.0	47.6	38.6	30.9		
	0.4%	WB	34.0	34.0	45.3	60.2	66.6	72.0	73.1	72.8	66.8	57.8	49.2	35.8		
		MCDB	34.3	34.4	52.1	77.4	82.4	89.0	87.7	89.7	84.7	68.0	51.0	36.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	31.9	32.5	38.2	57.2	63.9	69.3	70.8	69.0	62.1	53.7	44.7	33.6		
		MCDB	32.2	33.0	40.9	72.6	80.6	86.4	85.9	85.5	77.6	63.9	46.6	34.1		
	5%	WB	29.8	30.8	35.4	54.2	61.9	67.5	69.2	67.1	60.0	51.0	41.3	32.3		
		MCDB	30.5	31.7	37.1	66.5	77.6	84.1	84.3	84.2	74.0	59.5	43.1	32.9		
	10%	WB	26.6	27.8	33.8	51.3	60.0	65.7	67.6	65.5	58.0	48.6	38.6	30.6		
		MCDB	27.5	28.8	35.2	62.1	74.1	80.9	82.1	82.3	71.1	55.7	40.2	31.5		
	5% DB	MDBR	14.0	16.2	15.4	21.1	24.3	24.4	23.9	25.3	23.8	18.1	11.8	13.1		
		MCDBR	9.6	10.7	14.2	29.2	30.6	30.3	28.6	31.0	32.0	28.0	14.1	9.6		
		MCWBR	9.8	10.5	10.8	15.1	13.8	12.2	11.1	11.5	15.4	15.8	11.6	9.0		
		MCWBR	9.4	10.3	13.1	26.5	27.9	26.5	25.1	27.5	28.3	24.0	12.3	9.8		
(35) Mean Daily Temperature Range	5% WB	MCWBR	9.7	10.4	10.4	14.7	14.1	11.8	10.9	11.9	14.6	14.7	10.9	9.6		
	Clear-Sky Solar Irradiance	taub	0.303	0.330	0.382	0.431	0.411	0.403	0.411	0.429	0.380	0.354	0.334	0.305		
		taud	2.067	2.010	1.960	2.078	2.238	2.322	2.303	2.228	2.328	2.316	2.216	2.082		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	210	239	249	254	267	271	266	253	253	235	200	186		
		Edh at Noon	25	36	46	46	42	38	39	40	32	27	22	21		
(44) All-Sky Solar Radiation	RadStd		377	743	1175	1473	1856	2044	1984	1654	1164	607	278	235		
			48	55	134	137	154	180	136	124	123	96	44	51		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+1.84	-1.10	-1.21	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	+1.98	N/A	-1.03	N/A	N/A	N/A	N/A

Nomenclature: See separate page